

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CT
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INDEX
NUMBER

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BIOLOGY

9648/02

16 September 2016

Paper 2 Core Paper

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

All working for numerical answers must be shown.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, fasten all your work securely together in 2 separate section A and B.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
8	
Section B	20
9 / 10	
Total	

This document consists of **26** printed pages including the cover page.

[Turn Over

Section A

Answer **all** questions in this section.

- 1 Fig 1.1 shows some plant cells undergoing mitosis.

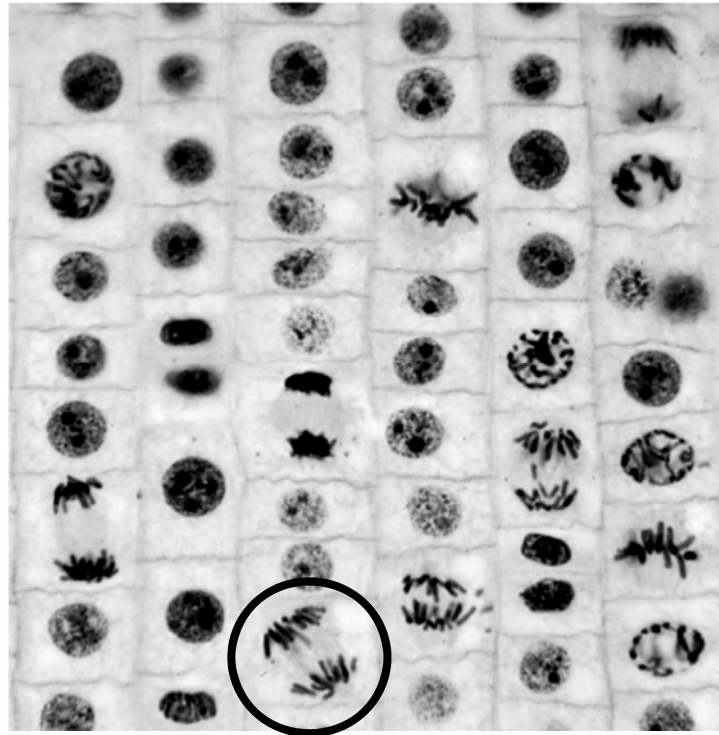


Fig. 1.1

Fig. 1.2 shows the changes in amount of DNA at different stages of the plant life cycle.

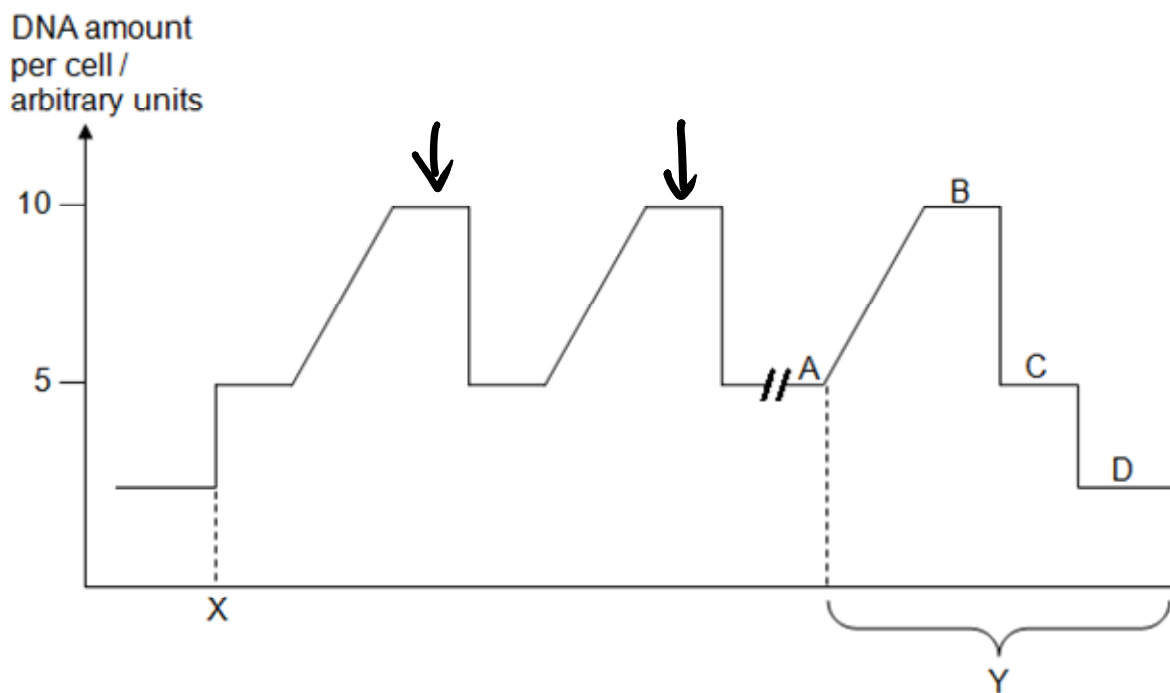


Fig. 1.2

- (a) Mark out with an arrow ↓ clearly on Fig 1.2 which part of the graph corresponds to the stage circled in Fig. 1.1. [1]

Accept all part of line except the corners (accept one arrow on either plateau, but not on the Meiosis plateau)

- (b) From stages **A** to **D** in Fig. 1.2, state all stages

- (i) that has/have the same number of chromosomes as shown in Fig. 1.1; [1]

A and B;;

- (ii) that has/have the **different** number of chromosomes as shown in Fig. 1.1; [1]

C and D;;

- (c) Explain the significance of the stages in **Y** in genetic variation. [4]

- a) Crossing over between non-sister chromatids of homologous chromosomes/bivalents/homologous pair takes place during prophase I*;;

OR

where equivalent portions of non-sister chromatids of homologous chromosomes break and rejoin during prophase I;;

- b) gives rise to new combination of alleles / mixing of alleles from both parental chromosomes which creates genetic variation in gametes;;

A: new linkage groups in place of new combination of alleles

- c) Independent assortment of homologous chromosomes/bivalents/homologous pair at metaphase plate during metaphase I and their subsequent separation during anaphase I;;

OR

Homologous chromosomes are arranged independently of other homologous pairs at metaphase plate during metaphase I and their subsequent separation during anaphase I;;

- d) results in 2ⁿ possible (types of) gametes where n is the number of homologous pairs;;

OR

Gametes with different combinations of parental (maternal and paternal) chromosomes;;

- (d) Explain the significance of the event occurring at **X**. [1]

- a) X refers to fertilisation*; Or random fusion of gametes* results in

- b) greater variation/varied offspring with different genotypes and phenotypes;

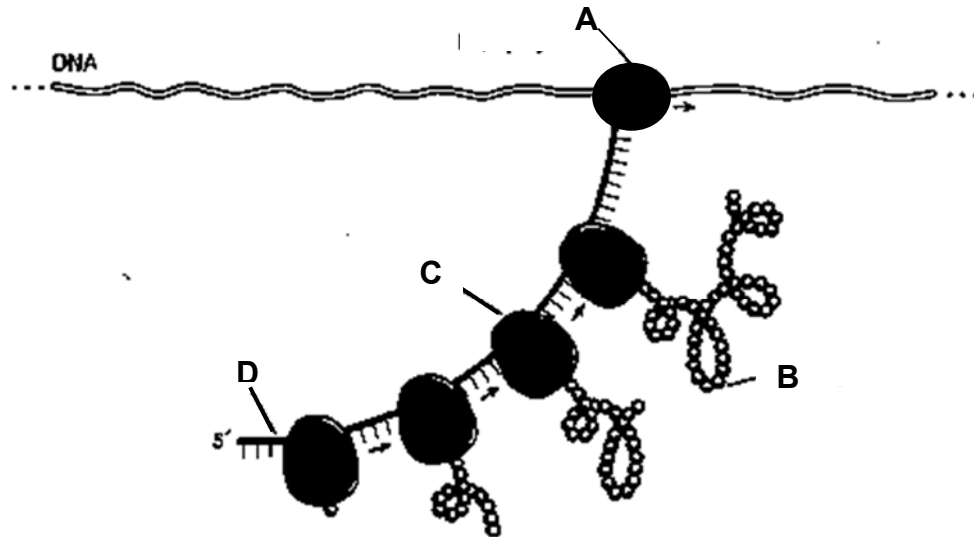
OR

Restoration of the diploid number of chromosomes;

[Total: 8]

[Turn Over

2 Fig. 2.1 shows a bacterial cell during protein synthesis.



(a) Identify A – D. [2]

A - RNA polymerase
B- growing polypeptide chain
C - ribosome
D - mRNA

(b) With reference to Fig. 2.1, suggest evidences that indicate that these processes occurred in a prokaryotic cell. [1]

- a) Presence of polyribosomes/ polysomes;**
 - b) Transcription and translation occur simultaneously;**
 - c) No membrane separates transcription and translation; (OWTTE).**
- Any 2**

(c) Briefly describe how structure A differs from structure C.[2]

Structure C is ribosome, Structure A is RNA polymerase

- a) Ribosome composed of ribosomal RNA and proteins while RNA polymerase is a protein made up of amino acids;;**
- b) RNA polymerase is single unit while ribosome consists of 2 subunits (1 large and 1 small ribosomal subunit);;**
- c) Ribosomes have 3 binding sites (APE) while RNA polymerase has 1 active site present;;**

(d) Activated tRNA can be seen entering structure C. Explain how the correct amino acid is joined to a tRNA. [4]

- The process that joins the correct amino acid to the tRNA is known as amino acid activation.
- Attachment of amino acid to a specific tRNA is catalysed by an enzyme called aminoacyl tRNA synthetase
- These enzymes have active sites which will recognize and fit only a specific combination of amino acid and anticodon of a tRNA that are complementary to the active sites
- As there are 20 commonly found amino acids, there will be at least 20 different aminoacyl tRNA synthetases.
- The synthetase enzyme catalyses the covalent attachment of the amino acid to its tRNA
- In a energy releasing process driven by hydrolysis of ATP.
- The resulting aminoacyl tRNA complex is released from the enzyme and will deliver its amino acid to the growing polypeptide chain on a ribosome.
- ensures that correct amino acid as specified by the genetic code is matched to the correct tRNAs

[Total: 9]

- 3 (a) Fig. 3.1 shows the viral load in the blood and CD4 T cell counts over the course of a typical HIV infection.

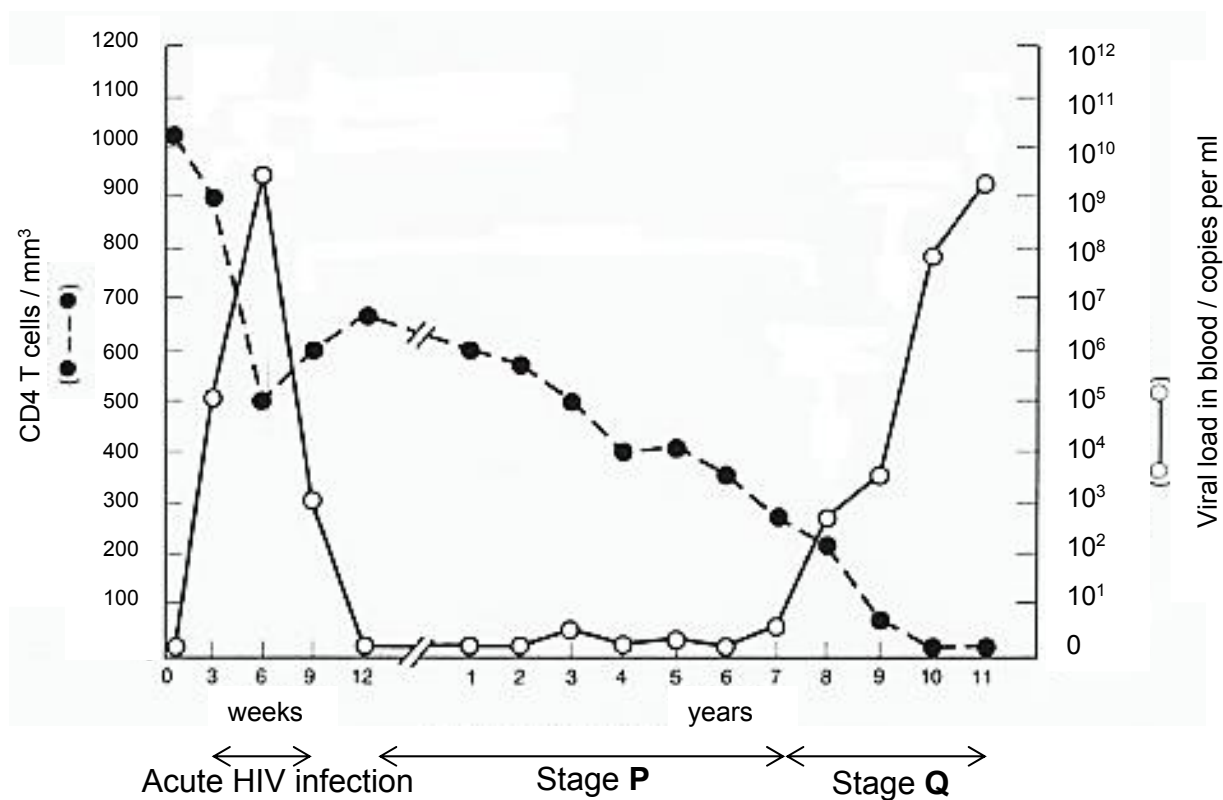


Fig. 3.1

(i) Many cells in Stage **P** were found to contain viral DNA. Explain the significance of Stage **P**. [2]

- a) Double stranded DNA reverse transcribed from the viral RNA ;
- b) integrates into the host cell DNA via integrase ;
- c) So that the viral DNA (encoding viral proteins) can replicate along together with the host cell DNA;
- d) Idea of results in production of more virions that carries the viral DNA (hence more viral progeny can be produced upon activation of the host cells) ;

(ii) In an experiment, it was found that the concentration of the enzyme, histone deacetylase (HDAC) in infected cells at Stage **P** was higher than in infected cells not at Stage **P**.

Suggest an explanation for the presence of abnormally high levels of HDAC in infected cells at Stage **P**. [2]

- a) Deacetylation of lysine residues in histone tails become positively charged
- b) results in increased affinity of the histone complex for the DNA molecule. Histone deacetylase cause chromatin to be more condensed / compacted/ packed more tighter
- c) Prevents transcription and translation of viral proteins during stage P
- d) Viral proteins not present on infected cells → not detected and destroyed by immune system;

(iii) With reference to Fig. 3.1, describe and explain the changes in the amount of HIV proteins and CD4+ T-cells count during Stage **Q**. [2]

- a) From 7 to 11 years after infection, viral proteins increases from 5 copies per ml to $10^9/10^{10}$ copies per ml because viral proteins are actively transcribed from integrated HIV DNA/provirus to form new viruses;;
- b) CD4 T cells decrease from 300 cells per mm^3 to 10 cells per mm^3 as budding of large amount of virions from the cell surface of CD4 T cells may disrupt the cell membrane sufficiently for cell to die / Hijacking of cellular machinery and resources towards producing new virions disrupts normal activities needed for cell survival, eventually causing cell death.;;

(iv) Suggest how viral load may be measured in the blood. [1]

- a) Via detection of HIV RNA in blood sample;; OR
- b) Via detection of viral proteins ;;

(b) HIV enters a CD4 cell. Gp120 binds to CD4 receptor on host cell surface membrane. Gp120 undergoes conformational change enabling gp120 to bind to co-receptors CCR5 on host cell membrane. Interaction between gp120 and co-receptor brings about conformational change to gp41 on viral envelop leads to fusion of HIV envelope to the host cell surface membrane, gaining entry.

Fig. 3.2 shows a CCR5 co-receptor, which is a protein.

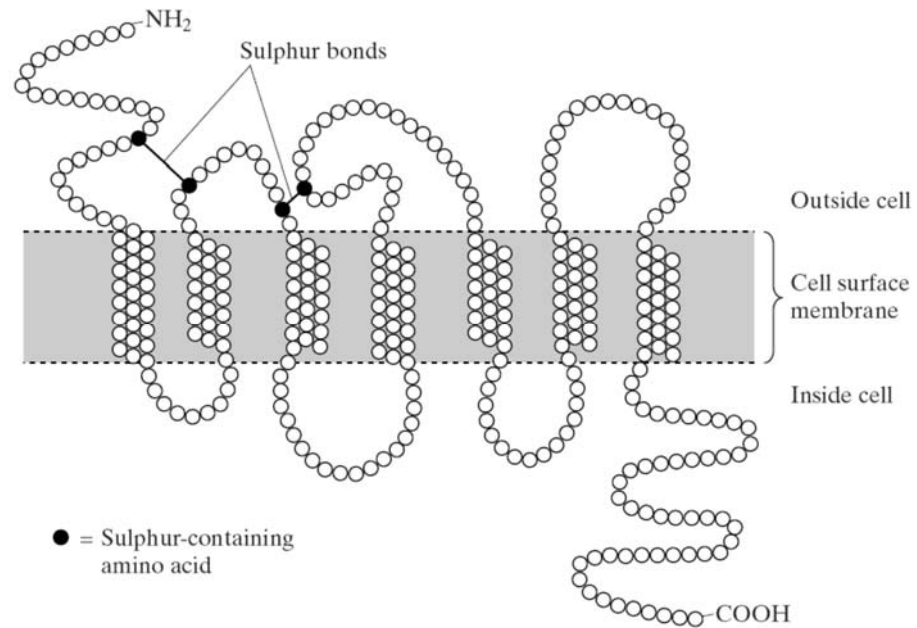


Fig. 3.2

Using information from Fig. 3.2, describe the protein structure of the CCR5 co-receptor. [2]

- a) **Primary Structure – sequence of amino acid in polypeptide chain;**
 - b) **Secondary structure consist of folding/helix in membrane involving mainly H-bonds between carbonyl and amino group;**
 - c) **Tertiary structure – Loops outside membrane and involves R group interactions;**
 - d) **Role of disulphide bonds present to maintain the tertiary structure of the protein;**
- (c) Integrase is one of the key targets for anti-retroviral therapy. An example of an anti-retroviral drug that acts on integrase is raltegravir. Fig. 3.3 shows the normal reaction catalysed by integrase, and Fig. 3.4 shows the effect of raltegravir on integrase.

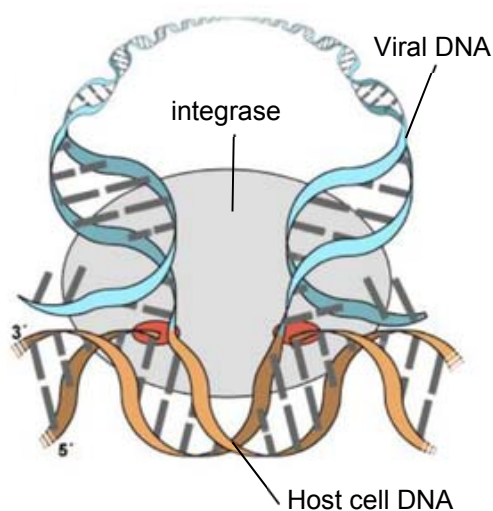


Fig. 3.3

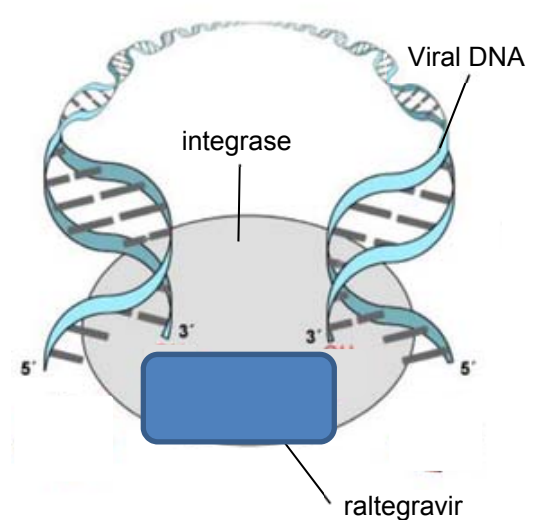


Fig. 3.4

Based on Fig. 3.3 and Fig. 3.4, describe the mode of action of raltegravir on integrase. [2]

- a) Raltegravir is a competitive inhibitor;
- b) It binds to the active site of integrase;
- c) Prevents integrase from binding to dsDNA;
- d) double stranded DNA will not be incorporated into the host cell's DNA as a provirus (by integrase);

[Total: 11]

- 4 "Cancer is a disease of the genome, triggered by the accumulation of genetic errors that eventually transform a normal cell into a tumour cell. Such mutations might inactivate genes that normally oppose tumour development, or activate genes that drive cell growth or interfere with cell differentiation or death.

This leads to the 2 classes of genes which are responsible for cancer formation."

Nature **417**, 906-907 (June 2002)

"Cancer is largely a disease of older people."

Nature Reviews Cancer **5**, 655-662 (August 2005)

- (a) Using the above information, explain why cancer usually occurs in older individuals [2]

- a) a single mutation in the cell is not enough to cause the cell to be cancerous as cancer development is a multi-step process;
- b) Time is required to accumulate mutations in many genes in the same cell for it to become fully cancerous;
- c) Older individuals are exposed longer to mutagens over their lifetime;
- d) Increasing the chance/probability of mutations occurring;

- (b) With specific named examples, distinguish between the 2 classes of genes stated in **line 5**. [3]

1 mark for both correct named examples
Any 2 differences

	Proto-oncogene (e.g. Ras gene)	Tumor Suppressor Gene (e.g. p53)
a) Primary function of gene product	<u>stimulate normal</u> cell division	<u>prevent uncontrolled cell</u> division
b) How does it attribute to the occurrence of cancer?	<u>Over-activation</u> (turn on) of proto-oncogenes → oncogene → occurrence of cancer	<u>Inactivation</u> (turn off) of TSG → occurrence of cancer
c) Type of mutation and its consequence	<u>Gain of function</u> mutation → <u>increase</u> in the amount of gene product/ increase in the intrinsic activity of the gene product <u>A mutation in only one allele of</u>	<u>Loss of function</u> mutation → <u>decrease</u> in the amount/elimination of gene product, decrease in the activity of the gene products <u>Mutations in both alleles of a</u>

	a proto-oncogene, converting to an oncogene, is needed for the cell to lose growth control	TSG are needed for the cell to lose growth control.
d) Is inheritance involved?	Majority of oncogenes develop from mutations in normal genes (proto-oncogenes) <u>during the life of the individual</u> →somatic mutations	Abnormalities of tumor suppressor genes can be <u>inherited</u> as well as <u>acquired</u> <u>during the life of the individual</u> . →Germline and somatic mutation.

Fig. 4.1 illustrates the RAS–BRAf–MAPK (mitogen-activated protein kinase) signalling pathway, to control cell proliferation or differentiation by changes in gene expression. Signal-transduction genes such as Braf and those of the Ras family, which encode components of the MAPK pathway, are frequently mutated in cancers.

Extracellular signals (growth factors) that activate one of two types of receptor — receptor tyrosine kinases and G-protein-coupled receptors — can result in the activation of Ras, leading to activation of Braf and the downstream cascade

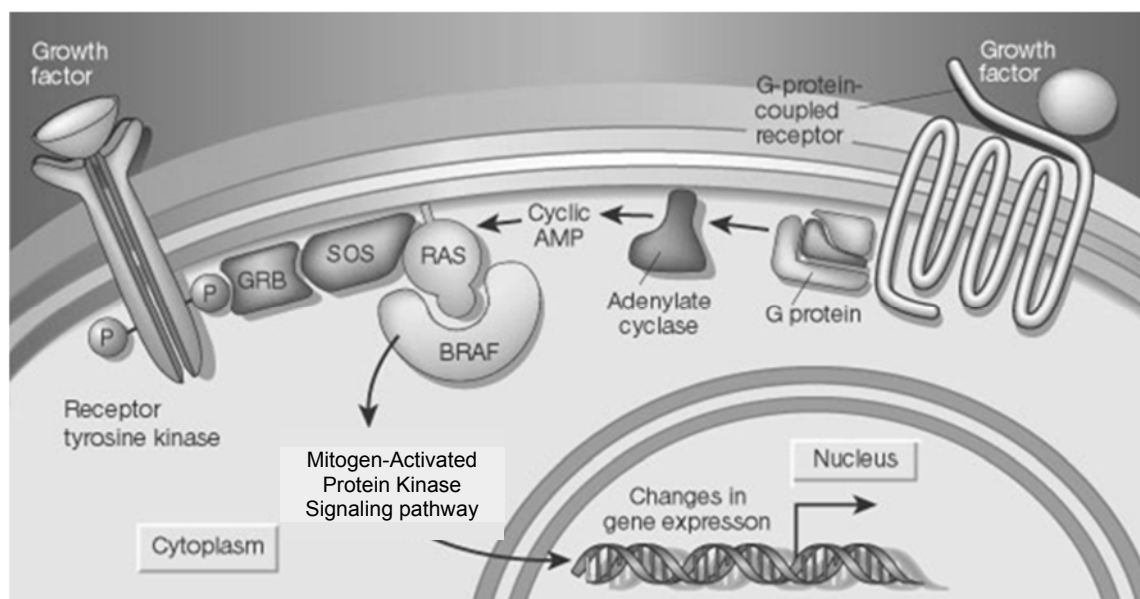


Fig. 4.1

(c) With reference to Fig. 4.1,

(i) Describe how Ras protein can be activated by **one** of the growth factors. [4]

- Growth factor (ligand) binds to G-protein-coupled receptors, receptor undergoes change in conformation which binds to inactive G-protein;;
- Causes a GTP to displace GDP on the G protein, activating it;;
- Activated G protein binds to adenylate cyclase, activating it;;
- Activated adenylate cyclase will catalyse the formation of cAMP from ATP, cAMP will trigger activation of Ras protein;;

OR

- e) Binding of growth factor causes two receptor polypeptides to associate, forming a dimer;;
- f) Dimerisation activates the tyrosine kinase parts of both polypeptides → each add phosphates to tyrosines on the tail of the other polypeptide/resulting in autophosphorylation of the tyrosines on the tail of the other polypeptide;;
- g) Activated tyrosine kinase is recognised by relay protein GRB which binds to phosphorylated tyrosine and undergoes structural change that activates it;;
- h) Activated GRB will in turn allow SOS to bind and be activated, which will in turn bind to Ras protein and activate it;;

(ii) Describe how Ras may contribute towards the development of cancer. [3]

- a) Single point-mutation of the Ras proto-oncogene could potentially lead to the production of a hyperactive Ras protein which is active even when growth factors are absent;;
- b) Leading to an overactivation of Braf protein which could then overstimulate the mitogen-activated protein kinase signalling pathway;;
- c) This may then lead to an increase in gene expression of growth factors/proteins which stimulate cell proliferation;;

[Total: 12]

5 Fig. 5.1 shows the electron micrograph of a chloroplast with structures X, Y and Z labelled.

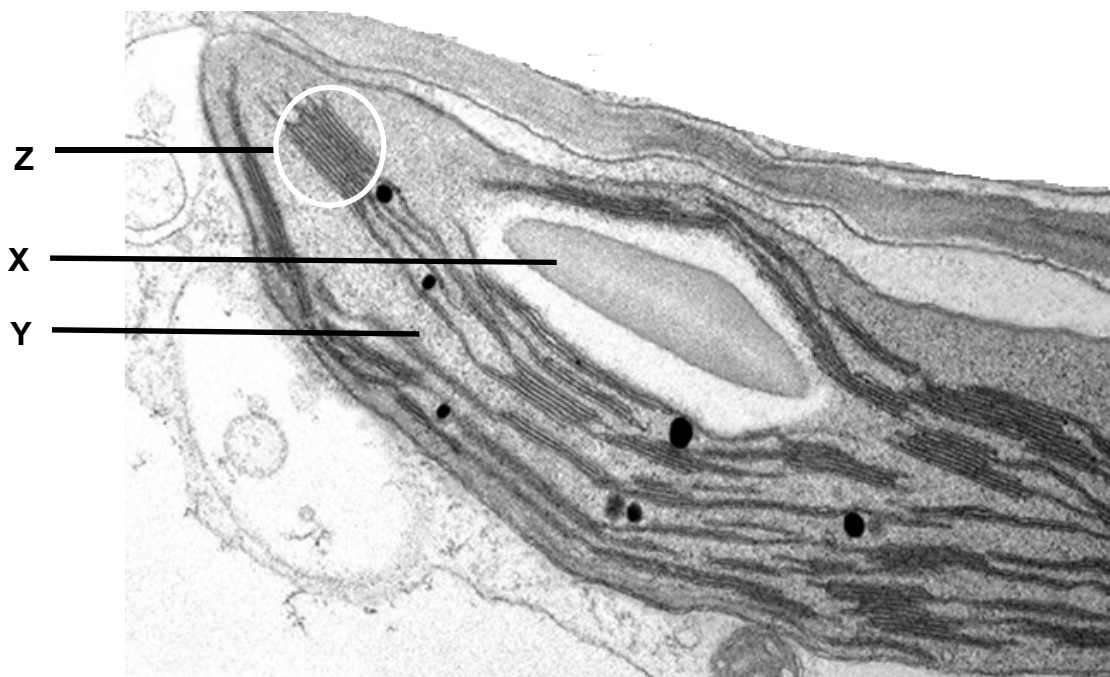


Fig. 5.1

(a) (i) Identify structures X and Y. [2]

X : Starch grain/granule

Y : stroma

(ii) Explain how the Z is adapted to carry out photophosphorylation. [2]

a. Z, is made from a stack of thylakoid membranes, / provides a large surface area;;

- b. Allows for embedding of many photosynthetic pigments / photosystems / light harvesting complexes, for light, harvesting / absorption;;
- c. The thylakoid membrane is impermeable to H^+ thus protons can only diffuse through stalked particles embedded in the membrane down its concentration gradient;;

(b) Rubisco is an important enzyme responsible for carbon fixation in the Calvin cycle. As Rubisco has an optimum pH of 9, it is most active during the daytime when the pH in the stroma is high.

Explain why Rubisco is most active during the daytime. [2]

- a. In the daytime, presence of light cause photophosphorylation to occur;;
- b. energy released from ETC is used to pump H^+ / actively transport H^+ from stroma into thylakoid space;;
- c. (low H^+ concentration in the stroma)→ reference to active site: the contact residues and/or catalytic residues in the active site are of the correct charge to bind the substrate molecules and catalyse the reaction respectively;;
- d. OR Ionic bonds between R-groups are stabilized confirmation of active site (most complementary to substrate

e.

A process known as photorespiration also takes place in photosynthetic cells. In this process, oxygen competes with carbon dioxide for the active site of the enzyme Rubisco.

Fig. 5.2 outlines the process of photorespiration.

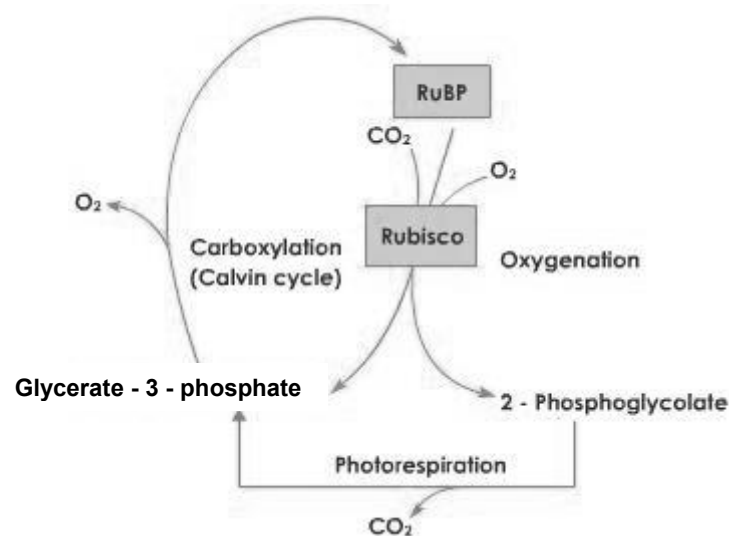


Fig. 5.2

(c) (i) Describe and explain the effects of an increase in oxygen concentration on photosynthesis. [3]

- a. An increase in oxygen concentration reduces the rate of photosynthesis / increases the rate of photorespiration;;
- b. This is less Rubisco is available for CO₂ / more oxygen competing with CO₂ for Rubisco;;
- c. This results in less CO₂ , fixation for Calvin cycle;;
- d. There is less, glycerate-3-phosphate, produced and less RuBP being, regenerated ;;

(iii) Suggest why the process outlined in Fig. 5.2 is known as photorespiration. [1]

- The process uses oxygen and, excretes / produces , carbon dioxide;
- Light energy / non-cyclic photophosphorylation / light dependent reaction / products of the light dependent reaction / ATP and NADPH, is required for this process to occur;
- The same photosynthetic enzyme / Rubisco is used and allows the Calvin cycle to continue;;

[Total: 10]

- 6 An inbred variety of maize, **A**, with finely striped leaves was found to have high resistance to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

The above F_1 generation was test cross with variety **A** and yielded the following results:

finely striped leaves and high resistance	80
finely striped leaves and low resistance	20
entirely green leaves and high resistance	22
entirely green leaves and low resistance	78

- (a) A chi-squared test was performed on the results of the cross to determine if the results of the test cross depart significantly from the expected ratio. Calculate the χ^2 value using the formula provided below. [1]

Formula for χ^2 calculation

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad v = c - 1$$

where

Σ = 'sum of...'

v = degrees of freedom

c = number of classes

O = observed 'value'

E = expected 'value'

Category	Observed number (O)	Expected ratio	Expected number (E)	$(O - E)^2$	$\frac{(O - E)^2}{E}$
finely striped leaves and high resistance	80	1	50	900	18
finely striped leaves and low resistance	20	1	50	900	18
entirely green leaves and high resistance	22	1	50	784	15.68

entirely green leaves and low resistance	78	1	50	784	15.68;
TOTAL	200				67.36 (2 d.p.);

Table 6 .1

degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.31	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (b) Using Table 6.1, explain the conclusion drawn from the χ^2 test to determine if the observed numbers conformed to the expected. [3]
- Since the calculated χ^2 value 67.36 more than critical value 7.82 at $p = 0.05$ at d.f.= 3, null hypothesis is rejected.
 - At 3 degree of freedom
 - Value of p is less than 0.001.;
 - The results of the χ^2 test suggest that there is a significant difference between the observed and the expected values(1:1:1:1).
 - Any difference is not due to chance alone but other factors e.g linked genes are at work.
 - Reject null hypothesis;
- (c) Using the following symbols:
- | | | | |
|----------|---------------------|----------|---------------------|
| G | entirely green leaf | g | finely striped leaf |
| R | low resistance | r | high resistance; |

Draw a genetic diagram to explain the difference between expected and actual results of the cross.

F₁ phenotype : entirely green leaves and low resistance x finely striped leaves and high resistance

F₁ genotype (test cross) : $\frac{G R}{g r}$ x $\frac{g r}{g r}$

Gametes formed : $\frac{GR}{g r}$ $\frac{Gr}{gR}$ x $\frac{gr}{gr}$
 Parental type recombinant type

Punnett square

	$\frac{GR}{g r}$	$\frac{gR}{g r}$	$\frac{Gr}{g r}$	$\frac{gr}{g r}$
$\frac{gr}{g r}$	$\frac{G R}{g r}$ entirely green, low resistance (Parental typed offspring)	$\frac{g R}{g r}$ striped leaves, low resistance (recombinant typed offspring)	$\frac{G r}{g r}$ entirely green, high resistance (recombinant typed offspring)	$\frac{g r}{g r}$ striped leaves, high resistance (parental typed offspring)

offspring Genotype : $\frac{G R}{g r}$ $\frac{g R}{g r}$ $\frac{G r}{g r}$ $\frac{g r}{g r}$

offspring Phenotype : entirely green, low resistance striped leaves, low resistance entirely green, high resistance striped leaves, high resistance

[4]
[Total:8]

7(a) Fig. 7.1 is an electron micrograph of a mitochondrion in a neurone.

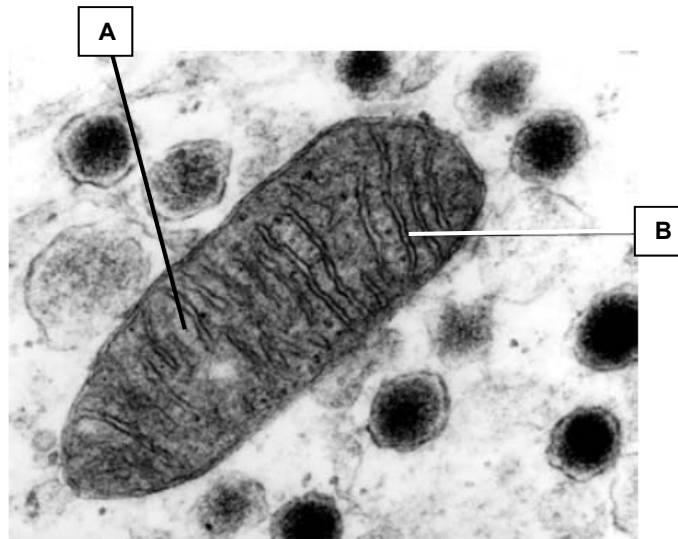


Fig. 7.1

Two stages of respiration occur in mitochondria. These are the Krebs cycle and oxidative phosphorylation.

- (i) Complete the table below by naming the structures labelled **A** and **B** and stating which of the stages of respiration occur in each.

	name of structure	stage of respiration
A	matrix	Kreb cycle
B	cristae / inner membrane	oxidative phosphorylation

- (ii) Explain the need for mitochondria along the axon in terms of nerve impulse conduction. [2]
[3]

Along the neurone:

- For production of ATP to release energy
- required to drive sodium potassium pumps
- Where 3 Na⁺ are pumped out of cells and 2 K⁺ are pumped into nerve cell; → impt for RP

At end of axon where synapse is found:

- For the active pumping of Ca²⁺ out of the synapses after nerve impulses have been conducted across synapses.
- To release energy for the transport of synaptic vesicles which carry materials for the synthesis of membrane & neurotransmitter substances;
- For vesicles containing neurotransmitters to release their contents via exocytosis into the synaptic cleft when impulses arrive at the synapses;

- (b) Serotonin is a neurotransmitter which is produced by certain neurons in the brain. One of its effects is to increase the activity of sensory neurones in the brain. It also usually improves a person's mood and keeps the person awake. Fig. 7.2 shows a synapse at which serotonin is the neurotransmitter.

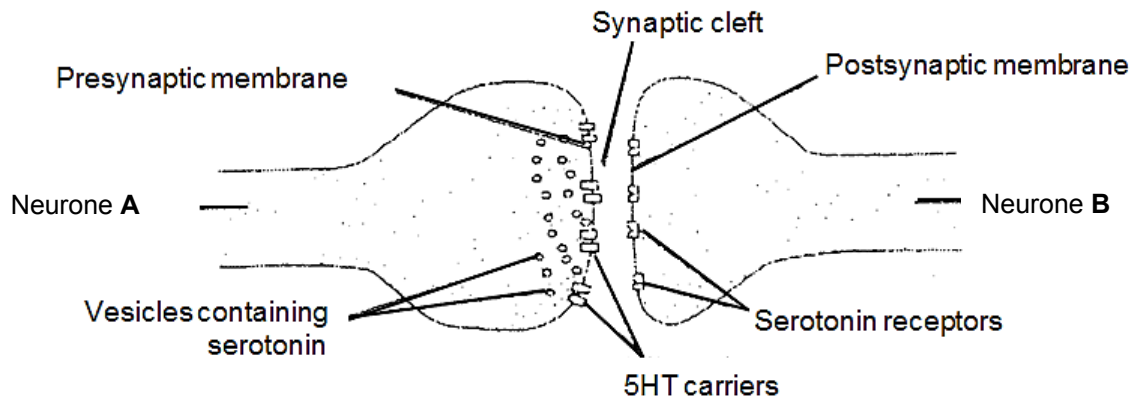


Fig. 7.2

- (i) Describe how the neurotransmitter is released into the synaptic cleft. [2]
- Arrival of nerve impulse/action potential
 - causes Ca^{2+} to enter/ influx of Ca^{2+} to cause
 - synaptic vesicles carrying the neurotransmitter, serotonin, move towards and fuse with the presynaptic membrane and
 - releases serotonin into synaptic cleft via exocytosis .
- (ii) Explain how the release of the neurotransmitter, serotonin, results in the transmission of an impulse from neuron A to neuron B. [2]
- Serotonin diffuses across the 20nm synaptic cleft post synaptic membrane
 - binding of serotonin to the receptors of ligand-gated Na^+ channel on postsynaptic membrane;
 - As a result, ligand gated sodium ion gates on membrane open. Sodium ions rush/influx of sodium into membrane down its electrochemical gradient
 - to depolarize the post-synaptic membrane to generate an EPSP.
- (iii) Suggest why nerve impulse can only travel in one direction across a synapse. [1]
- Voltage-gated Ca channels are found only on the pre-synaptic membrane, allowing translocation of neurotransmitter vesicles;
 - Neurotransmitter vesicles are only found in the pre-synaptic neuron;
 - Receptors for neurotransmitter are only found on the post-synaptic membrane;

[Total: 10]

- 8 There are over 40 Galapagos Islands including the small and isolated island named Daphne Major.

Studies were made every year from 1970 to 1989 on the beak size of the island's population of ground finch, *Geospiza fortis*, by measuring the beak length of every bird (Fig. 8.1). Larger finches with larger beaks are better at opening large seeds.



Fig. 8.1

From 1976 to 1978 there was a drought and only 15% of the ground finches survived and these did not breed during drought years. The most conspicuous feature of the survivors of the drought years was their large beak size.

During normal years, many drought intolerant grasses and herbs produce an abundance of small seeds. A few other drought-tolerant plants produce a much smaller number of large seeds which are not normally eaten.

- (a) Describe how environmental factors act as forces of natural selection, during drought years, on the beak size of finches. [4]
- Drought-tolerant plants which produced large seeds survived the drought better;;
 - The selection pressure being the type / availability of food source e.g. large seeds;;
 - Birds with larger beaks that can feed on large seeds are at selective advantage;;
 - can survive to maturity, mate, reproduce and pass on their favourable alleles (for larger beaks) to their offsprings;;
- (b) Suggest the role of the islands in the evolution of thirteen species of Darwin finches now found on the Galapagos Islands. [2]
- Geographical barrier / isolation between islands leads to reproductive isolation;;
 - Results in a lack of gene flow between population of finches;;
 - Different selection pressures on the different islands;;

Molecular analysis was carried out on the mitochondrial DNA (mtDNA) sequences of the Galapagos Islands finches and the Cocos finch found on the island of Cocos, 830 km to north-east of the Galapagos islands. Using mtDNA analysis data, a map showing the phylogeny of these finches was constructed as shown below in Fig. 8.2

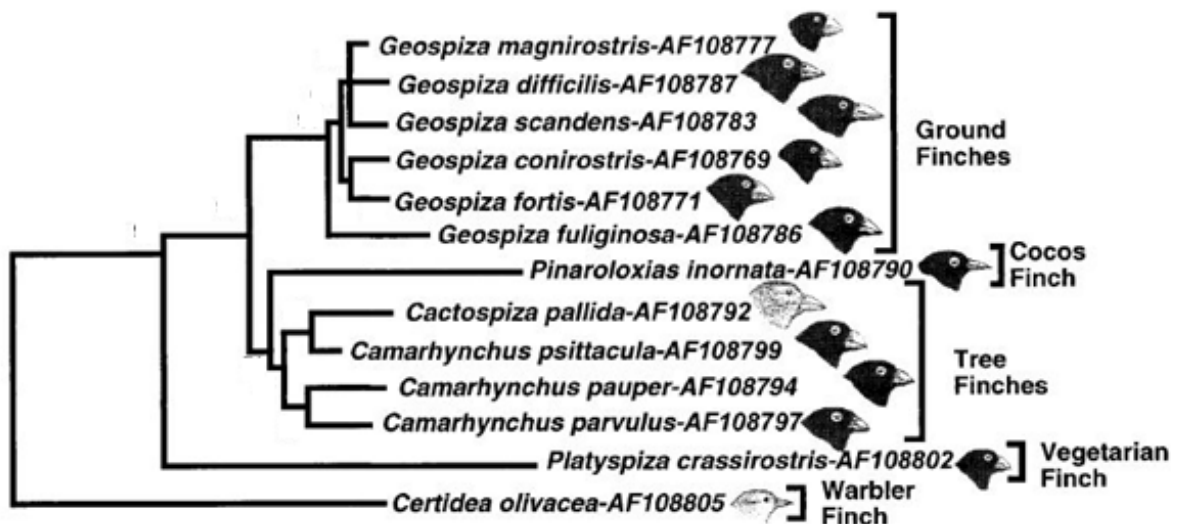


Fig. 8.2

- (c) Explain how DNA sequences can be used to determine evolutionary relatedness between species. [2]
- Compare regions of homology in the same gene e.g. cytochrome c gene / β -chain haemoglobin gene found in different species / compare homologous genes between species;;
 - The fewer the differences in DNA sequences of homologous genes between species, the more closely related the species are (vice-versa);;
- (d) Describe the advantages of using nucleotide data such as mtDNA in classifying organisms.[2]

OWTTE

- Using nucleotide data is unambiguous and objective. A, T, G, C are easily recognized and not confused with one another. They are not dependent on subjective judgements or observations involving quantitative differences;;
- Nucleotide data are quantifiable and can be converted to numerical form and open to statistical and mathematical analysis. This provides a quantitative tool for constructing phylogenetic trees with branch points defined by mutations in DNA sequence;;

Differences in the *cytochrome b* DNA sequence of several finches' species from Galapagos Islands and Island of Cocos were measured and plotted against time since divergence from the primitive ancestor (MYA) as seen in Fig. 8.3.

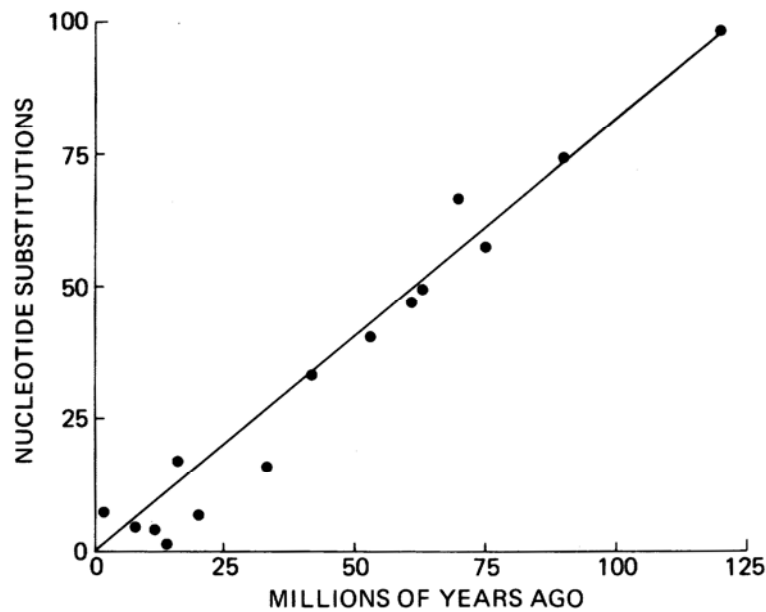


Fig 8.3

(e) Describe how these differences support the neutral theory of molecular evolution. [2]

- Changes in the nucleotide sequence arise through neutral mutation;
- for e.g. silent mutation or missense mutation where the change in amino acid does not occur in a critical region of the enzyme;
- There is no effect on the phenotype and fitness of organism/no selective advantage and thus allowed to accumulate;
- Small number of changes over millions of years indicating that rate of mutation is slow as cytochrome b gene is a crucial gene in living organisms;
- The plot of the line is straight, indicating that the rate of mutation is constant;

[Total: 12]

Section B

Answer **one** question.

Write your answers to this question on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 (a) Compare and contrast between glucagon and glycogen.

[4]

Compare (At least 1m)

- a) Both are macromolecules.
- b) Both are involved in maintaining blood glucose concentration.

Contrast

Points of comparison	Glucagon	Glycogen
c) Type of biomolecule	Protein	Carbohydrate / Polysaccharide
d) Site of production	Alpha cells of islets of Langerhans in pancreas	Liver cells, Muscle cells
e) Monomer	Amino acids	Alpha glucose molecules
f) Bonds between monomers	Peptide bonds	Alpha 1,4 glycosidic bonds, alpha 1,6 glycosidic bonds for branching
g) Structure adopted by each biomolecule	Tertiary structure – precise globular conformation	Helical structures with regular branching
h) Bonds stabilizing structure	Ionic bonds between charged R-groups, hydrogen bonds between polar R groups, disulfide bridges, and hydrophobic interactions	Hydrogen bonds between – OH groups projecting into the helix
i) Solubility in water	Soluble	Insoluble
j) Function	A hormone that raises blood glucose level	Energy storage molecules in animals

(b) Explain what is meant by primary secondary, tertiary and quaternary structure of a named protein.

[8]

R: collagen

- a) A protein with all 4 levels of structure is that of a globular protein, haemoglobin;
- b) Primary structure refers to the number, type and unique sequence of amino acids found;
- c) in each of the 4 polypeptide chains (2 alpha and 2 beta globin chains) linked by peptide bonds in a polypeptide chain;
- d) Secondary structure: segments of their polypeptide chain repeatedly coiled or folded;
- e) into some kind of geometrically regular secondary structure;
- f) The 2 main kinds are the α -helix and the β -pleated sheet;
- g) Both structures are stabilized by intrachain hydrogen bonds between the amino and carboxyl groups of the polypeptide backbone;
- h) The secondary structures then fold back on themselves to form a precise globular structure;
- i) Tertiary structure refers to the overall compact, globular, 3-D structure of each haemoglobin polypeptide chain;
- j) conformation of a polypeptide resulting from interactions between side-chains of the various amino acids in a polypeptide chain;
- k) The compact structure is maintained by four types of bonds, namely hydrogen bonds, ionic bonds, hydrophobic interactions as well as the stronger disulfide bonds;
- l) As the polypeptide folds into its functional conformation, amino acids with hydrophobic (non-polar) side chains congregate in clusters via hydrophobic interactions at the core of the protein, out of contact with water;
- m) Meanwhile, hydrogen bonds between polar side-chains and ionic bonds between positively and negatively charged side-chains also help stabilize tertiary structure;
- n) These are all weak interactions but their cumulative effect help to stabilize and give the protein a unique shape;
- o) Quaternary structure refers to the specific orientation of 2 or more polypeptide chains with respect to one another and the nature of interactions that stabilize this orientation;
- p) Each polypeptide chain in such a protein is called a subunit;
- q) There are 4 subunits for haemoglobin, namely 2 alpha and 2 beta polypeptide chains;
- r) The subunits are held together by hydrogen bonds, ionic bonds and hydrophobic interactions to form a multimeric functional protein;

(c) Describe the main properties of an enzyme and discuss their mode of action.

[8]

- a) Enzyme acts as a biological catalyst which speeds up the rate of chemical reaction by lowering activation energy;
- b) It remains unchanged at the end of reaction and it can be reused;
- c) They are globular proteins with the specific 3D conformation;
- d) and specific distribution of electrical charges at its active site;
- e) They highly specific for the substrates they recognise and the reactions they catalyse;
- f) Some enzymes recognise also one type of substrate (eg: catalase) or a specific functional group (eg: phosphate group) or type of chemical bond (eg: peptide bonds).
- g) Enzyme activity is affected by changes in pH, temperature, substrate and enzyme concentration and in the presence of cofactors and inhibitors;
- h) Enzymes are extremely efficient. Catalysed reactions are 10^3 to 10^8 times faster than uncatalysed reactions;
- i) Enzymes have high turnover number. The turnover number refers to the number of substrate molecules converted into products by one molecule of enzyme in one second;
- j) They are required only in small amounts;

- k) Precise active site to which the substrate molecules with complementary conformation and compatible chemical groups become attached;
- l) Collision between substrate and enzyme at the correct orientation causes the substrate molecules to bind to the enzyme molecule at its active site to form an enzyme-substrate (E-S) complex;
- m) Enzyme and substrate are held together by weak bonds, such as ionic, hydrogen and hydrophobic bonds, which can be made and broken rapidly;

Lock and Key hypothesis

- n) The conformation and chemical groups of the substrate is exactly complementary to the enzyme active site;
- o) Substrate binds to the active site by weak bonds such as hydrogen and ionic bonds to form an enzyme-substrate complex;
- p) It is then activated to form the products of the reaction, which no longer fit into the active site of the enzyme;
- q) Products are released into the surrounding medium;
- r) leaving the active site unchanged and free to receive other substrate molecules;

Induced-fit hypothesis

- s) Initially, the active site does not exist in a conformation that is exactly complementary to the substrate;
- t) As the substrate enters the active site, the amino acids which make up the active site are moulded or 'induced' into a precise shape complementary to the substrate;
- u) This brings a greater interaction between the chemical groups of the substrate and active site. (greater chemical and spatial compatibility);
- v) Enables the enzyme to perform its catalytic function more efficiently;
- w) For example, in reactions involving 2 or more reactants, the active site brings the substrates together in the proper orientation for the reaction to occur;
- x) As the active site clutches the substrates, the enzyme may stress the substrate molecules, stretching and bending critical chemical bonds within the molecule that must be broken during the reaction;
- y) The active site may also provide a microenvironment that is conducive to a particular type of reaction;

[Total: 20]

10 (a) With reference to the islets of Langerhans, describe what is meant by an endocrine gland. [7]

- a) Islets of Langerhans are clusters of cells in pancreas;
- b) A gland is a structure which secretes specific chemical substances called hormones;
- c) Each islet has a population of alpha cells which secrete glucagon and;
- d) a population of beta cells which secrete insulin;
- e) Endocrine gland is ductless instead;
- f) hormones are secreted into blood to be distributed throughout the body;
- g) transported to target cells/tissue;
- h) Thus, islets of Langerhans have a rich supply of blood vessels;
- i) Endocrine glands secrete hormones in small quantities;
- j) Endocrine glands are involved in homeostatic control/homeostasis that operates by negative feedback;
- k) E.g. islets of Langerhans control the secretion of hormones/ adjust hormonal output by monitoring/detecting blood glucose concentration;

@ 1m each

(b) Explain how the blood glucose concentration is regulated by insulin and glucagon. [8]

Regulation by insulin:

- a) Blood glucose level is normally maintained at ~90 mg/100 ml blood;
- b) High blood glucose (>90 mg/100 ml blood) is detected by beta cells of islets of Langerhans;
- c) Beta cells are stimulated to secrete insulin directly into the bloodstream;
- d) Insulin binds to the (insulin) receptors on target cells e.g. liver and muscle cells;
- e) After insulin-binding process induces dimerisation & autophosphorylation of tyrosine amino acids on the RTK,
- f) the activated receptor tyrosine kinase then activates a number of bound intracellular relay proteins.
- g) As insulin level rises,
 - i. increases rate of glycogenesis (= conversion of glucose to glycogen) in liver and muscle cells;
 - ii. increases rate of glucose uptake/permeability esp. by muscle cells;
 - iii. increases conversion of glucose into fats in liver cells i.e. fat deposition;
 - iv. Increase in the rate of amino acid absorption and protein synthesis;
 - v. increases rate of respiration of glucose instead of other energy source(glycolysis);
 - vi. inhibits/decreases gluconeogenesis (= conversion of proteins and fats into glucose)
 - vii. Decrease in glycogenolysis, ie. breakdown of glycogen to glucose in skeletal muscle and liver cells.
- (max 1m - any 2 pt)
- h) These mechanisms by insulin will lower blood glucose levels;
- i) When blood glucose level is restored to the normal set point/normal blood glucose concentration, beta cells detect and reduce insulin secretion.

Regulation by glucagon:

- j) Low blood glucose (< 90 mg/100 ml blood) is detected by alpha cells of islets of Langerhans;
- k) Alpha cells are stimulated to secrete glucagon directly into the bloodstream;
- l) Glucagon binds to the (glucagon) receptors on liver cells (only liver cells are glucagon's target i.e. only liver cells are sensitive to glucagon);
- m) to activate adenyl cyclase (enzyme) to form cAMP i.e. increases cAMP level
- n) which activates phosphorylase enzymes that catalyse glycogenolysis (= conversion of glycogen to glucose);
- o) As glucagon level rises,
 - i. Increase in glycogenolysis = stimulate breakdown of glycogen to glucose;
 - ii. Stimulates triglyceride breakdown in adipocytes.
 - iii. Inhibits glycogenesis in liver and skeletal muscle cells
 - iv. Increase in gluconeogenesis
- (max 1m - any 2 pt)
- p) Newly synthesized glucose leaves target cells into the bloodstream;
- q) These mechanisms by glucagon will increase blood glucose levels;
- r) Secretions of insulin and glucagon are regulated by negative feedback mechanism.

(c) Describe how the endocrine system differs from the nervous system. [5]

	Endocrine / hormonal system	Nervous system
a) Anatomical arrangement	Endocrine glands widely dispersed in your body and not structurally related to one another or to their target cells.	Specific structural arrangement betwn neurons and their target cells; interlinked → structural continuity in the system.

b) Type of information being transmitted and the messenger (s) involved	Relies on chemical transmission through the circulatory system; Information passes as a chemical messenger (hormone) thru blood stream. Hormones released into blood.	Involves both electrical & chemical transmissions along nerve fibres and between nerve fibres; information passes as electrical impulses along axons of neurons and as chemical messenger (neurotransmitters) across synapses. Neurotransmitters released into synaptic cleft, never into blood. Electrical signal via nerve fibres.
c) Distance of action of chemical messenger	Long distance (carried by bloodstream – always exert its effect on target organs away from the site of synthesis)	Very short distance (diffuses across synaptic cleft) - neurotransmitter
d) Speed of transmission	Slow rate (relies on diffusion through the bloodstream)	Rapid rate (due to saltatory conduction of nerve impulses at the nodes of Ranvier)
e) Speed of response	Slow acting (from minutes to hours)	Rapid / immediate (in milliseconds)
f) Duration of action	Long lasting (from minutes to days or longer)	Brief / short-lived (in milliseconds)
g) Effects	Diffused; usu. Widespread	Localized; very precise & exact
h) Major functions	Controls activities that require long duration rather than speed.	Coordinates rapid, precise responses; esp important in mediating interactions with the external environment.

[Total: 20]